

Name: _____

Homework #09: System Clock and SysTick Timer

SysTick Timer

1. The SysTick timer on the TM4C123G microcontroller is a ____-bit down-count timer.

Answer: _____

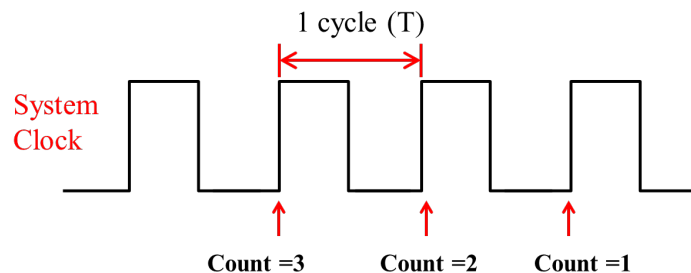
2. What is the maximum value that can be loaded into the SysTick timer? (Give your answer in hexadecimal.)

Answer: _____

3. If the SysTick timer reload register is loaded with the value 2, what is the counter value one clock cycle after reaching 0?

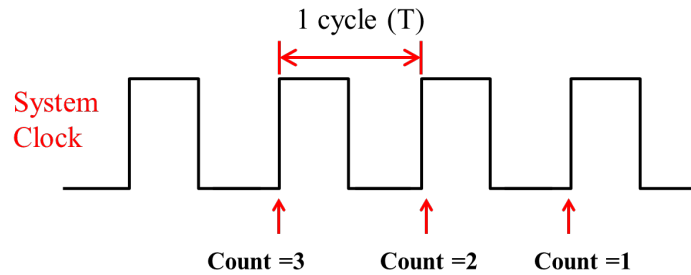
Answer: _____

4. If the SysTick timer is using the default 16 MHz system clock, how long does it take to count from 3 to 2 as shown in the image below. Give your answer in seconds using engineering notation.



Answer: _____

5. If the SysTick timer is configured to use the 80 MHz system clock, how long does it take to count from 3 to 2 as shown in the image below. Give your answer in seconds using engineering notation.



Answer: _____

6. What value would you place in the SysTick timer reload register if you wanted the count flag to go high after 10 periods of the clock. Give your answer as a decimal number.

Answer: _____

7. Write the assembly code to configure the SysTick timer to use the system clock. Do not overwrite any other settings. (*Do not use labels. Use the actual hex address of any configuration registers used.*)

Use the cod below to answer question 8-10

```
128 SysTick_Wait
129     LDR R1, =NVIC_ST_RELOAD_R
130     SUB R0, #1
131     STR R0, [R1]
132     LDR R1, =NVIC_ST_CURRENT_R
133     STR R0, [R1]
134     LDR R1, =NVIC_ST_CTRL_R
135
136 SysTick_Wait_Loop
137     LDR R3, [R1]
138     ANDS R3, R3, #_____ ;test COUNT bit to see if timer has hit 0
```

```
139    BEQ    SysTick_Wait_Loop
140    BX      LR
```

8. If the SysTick_Wait subroutine is sent the decimal value 100 in R0, the count flag will go high every _____ (decimal) clock cycles.

Answer: _____

9. What is the value of the SysTick Timer's current count right after line 133 is executed? (Note that line 133 is a write to the current value register.)

Answer: _____

10. What is the value of the mask (in hexadecimal) that should be used in the blank on line 138 that would correctly test for the SysTick Timer's count flag to go high and exit the subroutine if it has?

Answer: _____